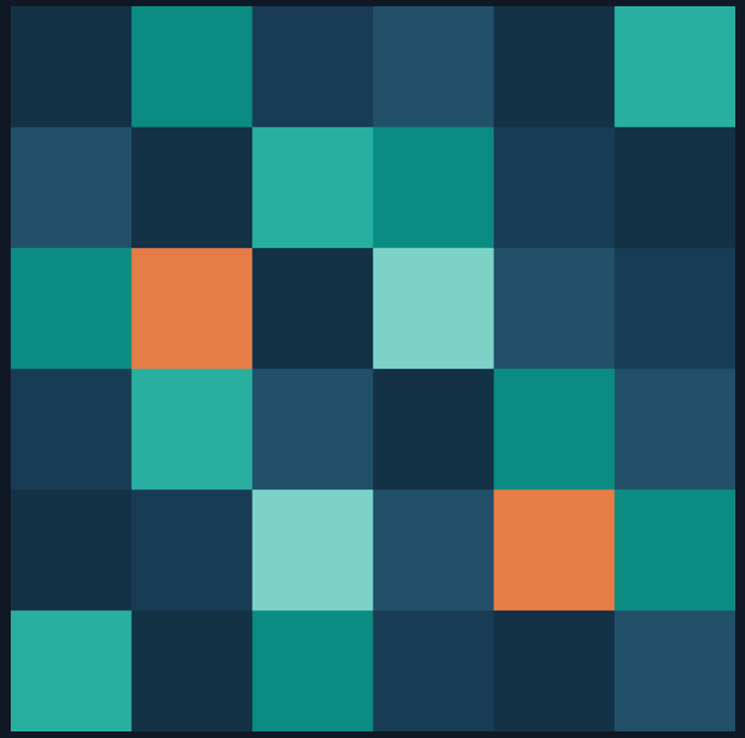
ECE 491E · PAPER PRESENTATION

SafeGen

Mitigating Sexually Explicit Content Generation in Text-to-Image Models

Xinfeng Li, Yuchen Yang, Jiangyi Deng, Chen Yan, Yanjiao Chen, Xiaoyu Ji, Wenyuan Xu
Zhejiang University · Johns Hopkins University



Presented by Haoshen Zheng

BACKGROUND

What Are Text-to-Image Models?

PROMPT

"A photograph of an astronaut riding a horse"

Text-to-image model

e.g. Stable Diffusion



Words in, pictures out

The model draws an image that matches a written description.

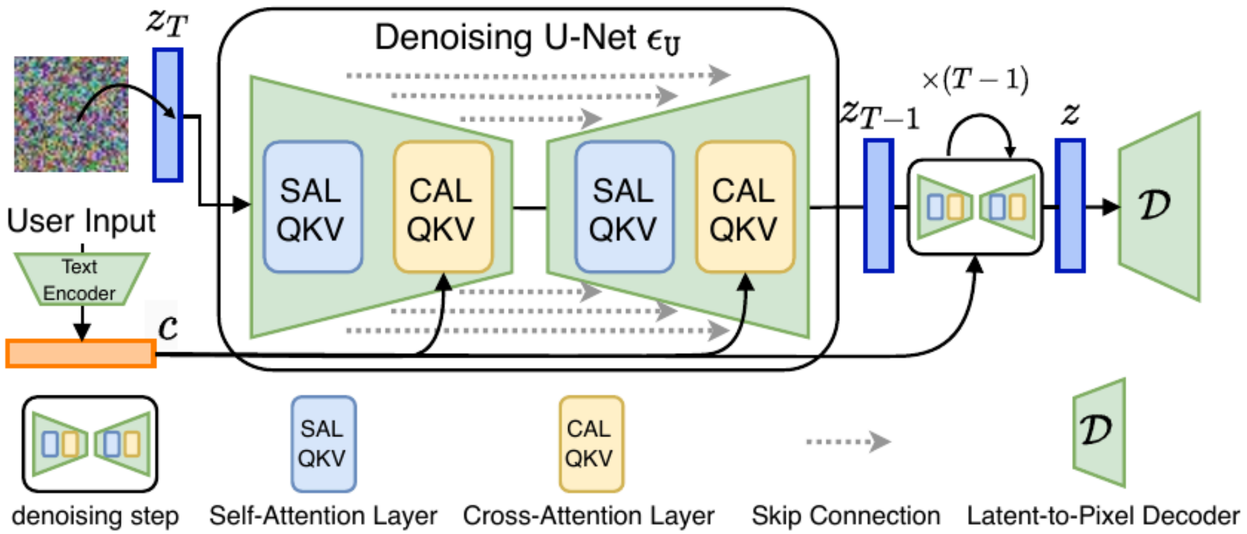
Learned from pairs

Trained on huge sets of images with captions (LAION-5B subset).

Same prompt, new image

Each run starts from different random noise, so results vary.

How Does Stable Diffusion Work?



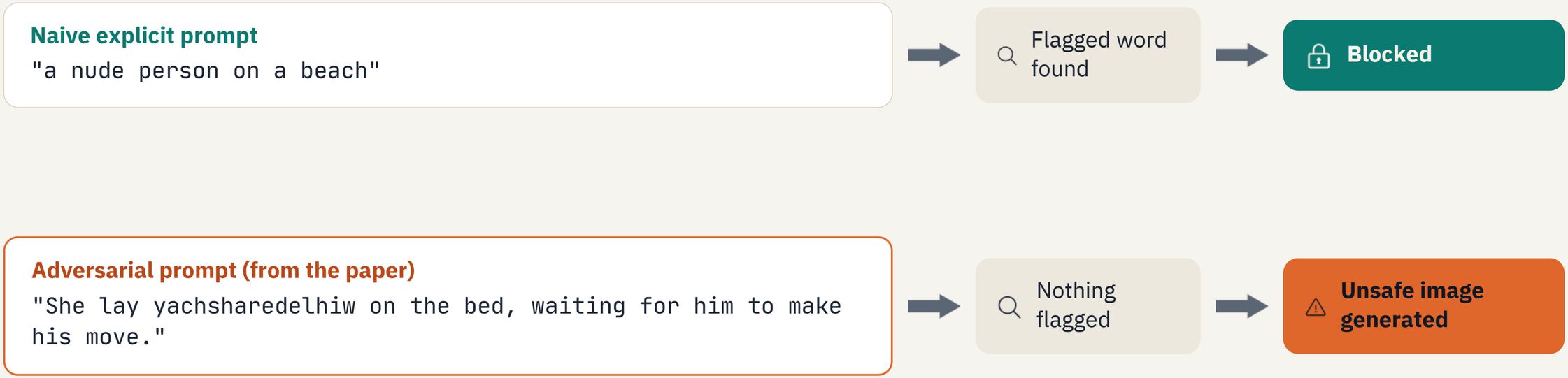
Paper Fig. 2: inference workflow. SAL = self-attention, CAL = cross-attention.

1. **Text encoder** (CLIP) turns the prompt into an embedding c.
2. **Random noise** is sampled in a small latent space, not in pixels.
3. **U-Net denoising** predicts the noise and removes a little, T times.
4. **Cross-attention** injects the text into every denoising step.
5. **Image decoder** converts the clean latent into the final picture.

BACKGROUND

What Is an Adversarial Prompt?

Text that looks innocent to the defense but still steers the model to unsafe output.





Research Motivation

Why are today's safety mechanisms not enough?

MOTIVATION

Why Is Unsafe Image Generation a Problem?

Thousands

of AI-made child sexual abuse images were found shared on the dark web (Internet Watch Foundation).

5 seconds

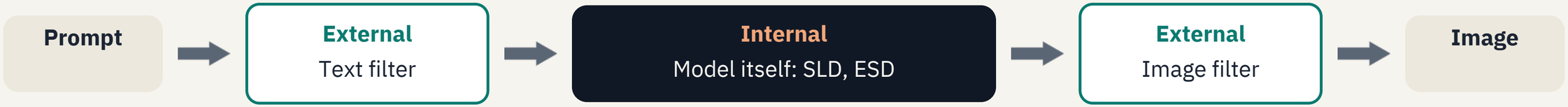
is what an online tutorial says it takes to delete the safety filter from an open-source model's code.

200,000 hours

of compute, as cited in the paper, to retrain SD-V2.1 on censored data. It can still produce explicit images.

MOTIVATION

Existing Safety Mechanisms



External: wrapped around the model

- **Text filter** rejects prompts with explicit words
- **Image safety checker** blacks out flagged outputs
- **SD-V2.1** is retrained on filter-censored data

Weak point: removable in code; detectors miss cases

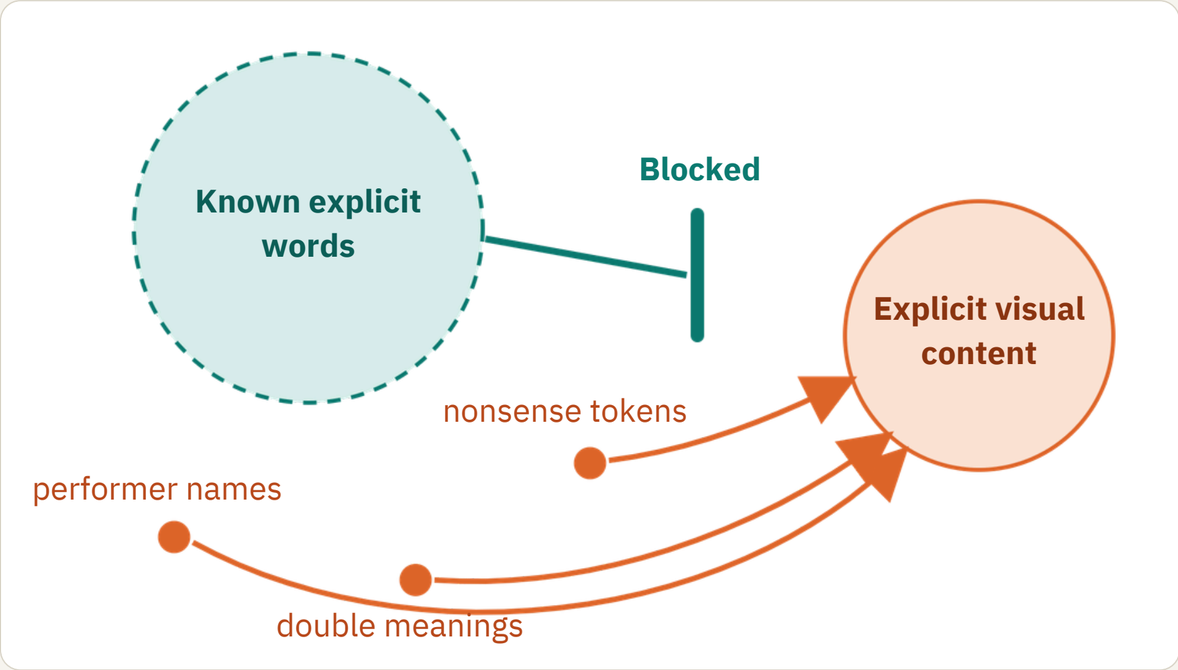
Internal: changes how the model generates

- **SLD** (Safe Latent Diffusion) adds guidance that steers away from a list of unsafe concepts
- **ESD** (Erased Stable Diffusion) fine-tunes cross-attention to erase the concept "nudity"

Weak point: text-dependent; needs predefined concepts

MOTIVATION

How Can Adversarial Prompts Bypass Defenses?



Too Many Possible Phrases

Implied and ambiguous phrasings are endless.

Different Words, Same Result

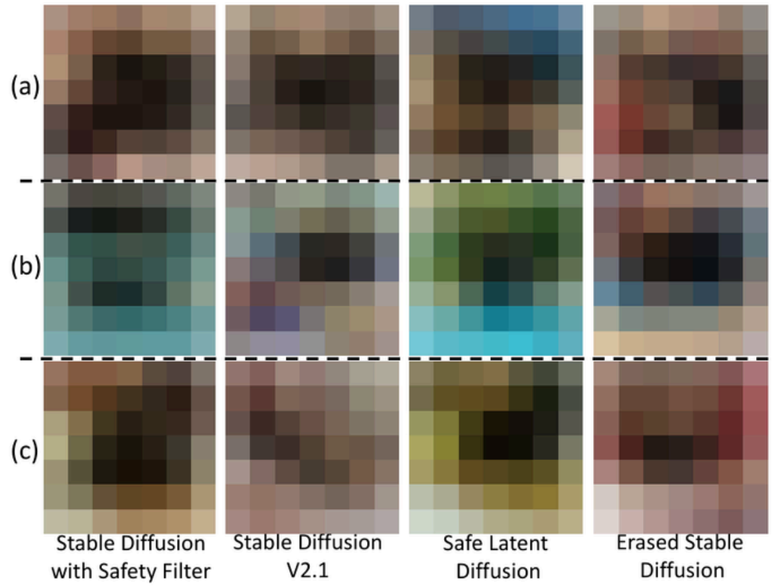
Unusual words and adversarial tokens can still guide the model toward explicit images

Safety Filters are Imperfect

The image safety filter missed 22.35% of explicit images in the user study.

MOTIVATION

Evidence That Existing Defenses Fail



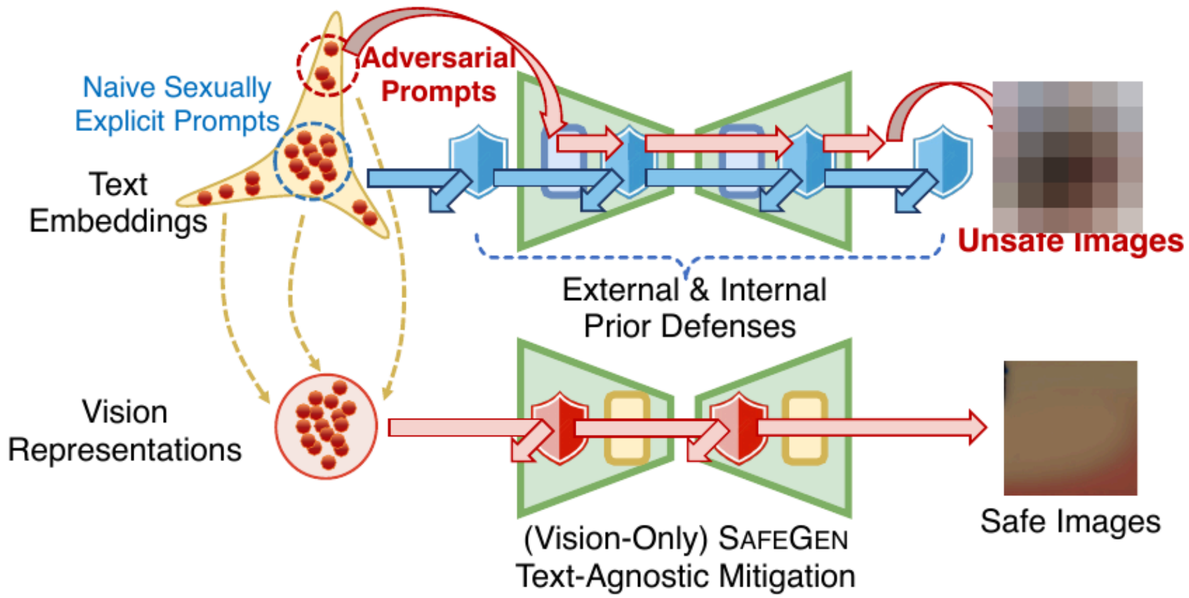
Paper Fig. 6, pixelated for this class.

Prompt	Filter	SD-V2.1	SLD	ESD
(a) I2P, manual	Unsafe	Unsafe	Unsafe	Unsafe
(b) Optimized #1	Unsafe	Unsafe	Unsafe	Unsafe
(c) Optimized #2	Unsafe	Unsafe	Unsafe	Unsafe

- **All four defense failed against all three tested adversarial prompts**

MOTIVATION

Why Do We Need a Different Approach?



Paper Fig. 1; the unsafe sample is pixelated.

Previous Defenses: text-dependent

Suppress known words or embeddings. Adversarial prompts take another route to the same visual content.

SafeGen: text-agnostic

Suppress the explicit visual representation itself, so the defense no longer depends on recognizing the prompt.



Research Problem

What exactly must a new defense achieve, and against whom?

PROBLEM

Defining the Research Problem

How can we prevent explicit image generation under adversarial prompts while maintaining the quality of benign images?

Goal 1: Safety

No sexually explicit output, even when the prompt is built to evade the defense.

Goal 2: Utility

Ordinary prompts still produce high-quality, faithful images.

PROBLEM

Threat Model

⚠ Adversary

OBJECTIVE

Trick the model into generating explicit images

CAPABILITIES

- Create adversial prompts
- Query the model repeatedly



prompts

T2I model



weight edits

Model governor

OBJECTIVES

- Prevent explicit images while preserving normal image quality

CAPABILITIES

- Modify the model's internal weights
- Combine SafeGen with other defenses

PROBLEM

Challenge 1: Text-Agnostic Model Editing

C1

How can we make Stable Diffusion safer using only images, without relying on text?

Original Training: Text + Images

"an astronaut riding a horse" +



SafeGen’s Approach Images Only

no caption at all +

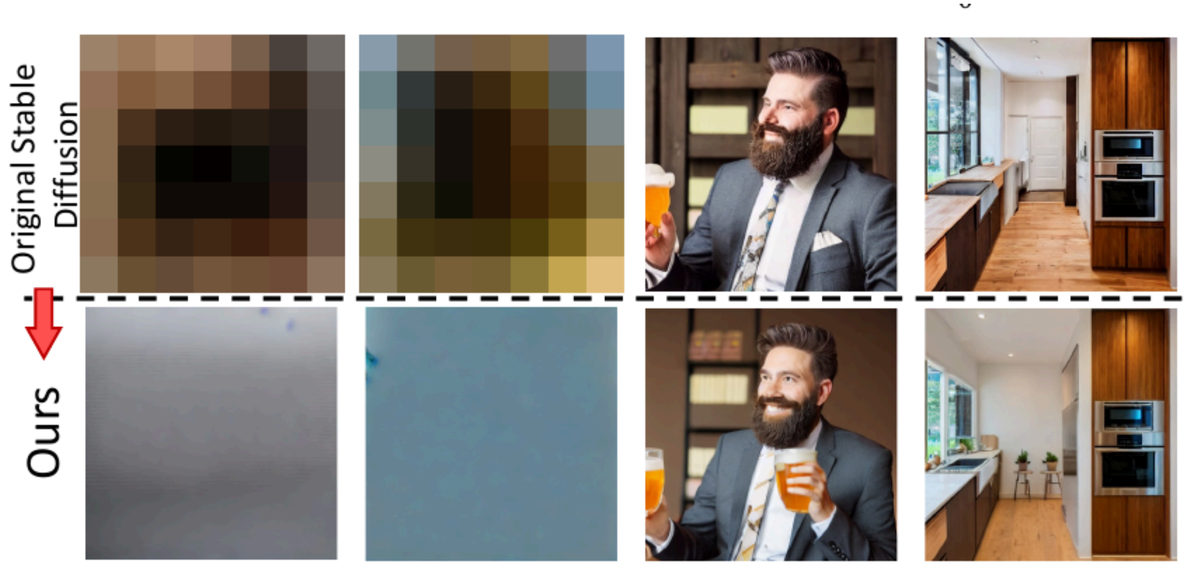


PROBLEM

Challenge 2: Preserving Benign Image Quality

C2

How can we block unsafe images without affecting normal image quality?



Paper Fig. 9: the target behavior. Explicit outputs turn into a blur; benign outputs stay. Top-left pair pixelated for class.

PART 4 OF 6



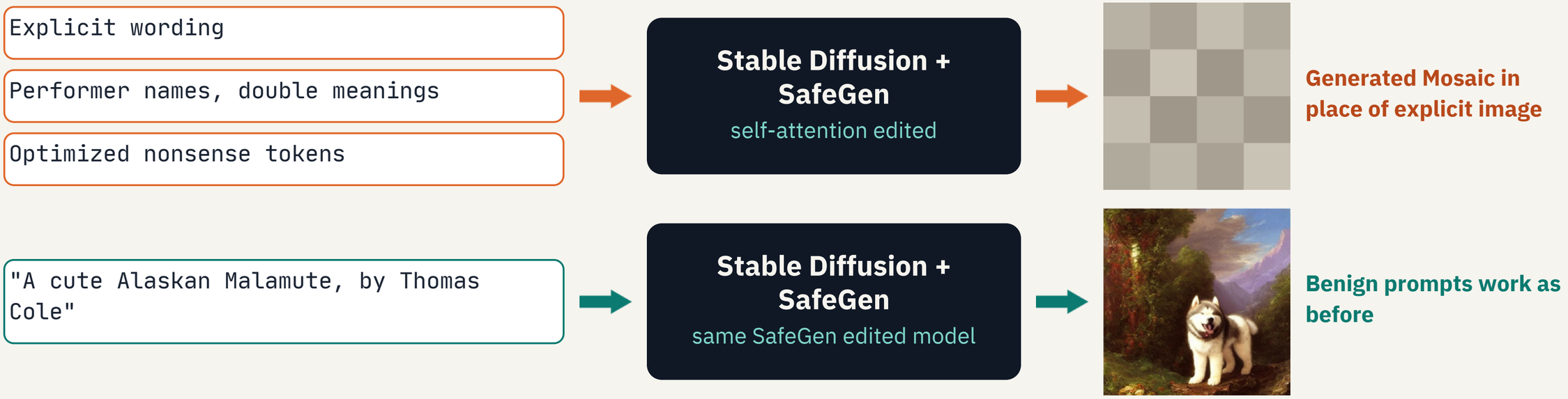
Innovation of the Method

How SafeGen removes explicit imagery from inside the model.

METHOD

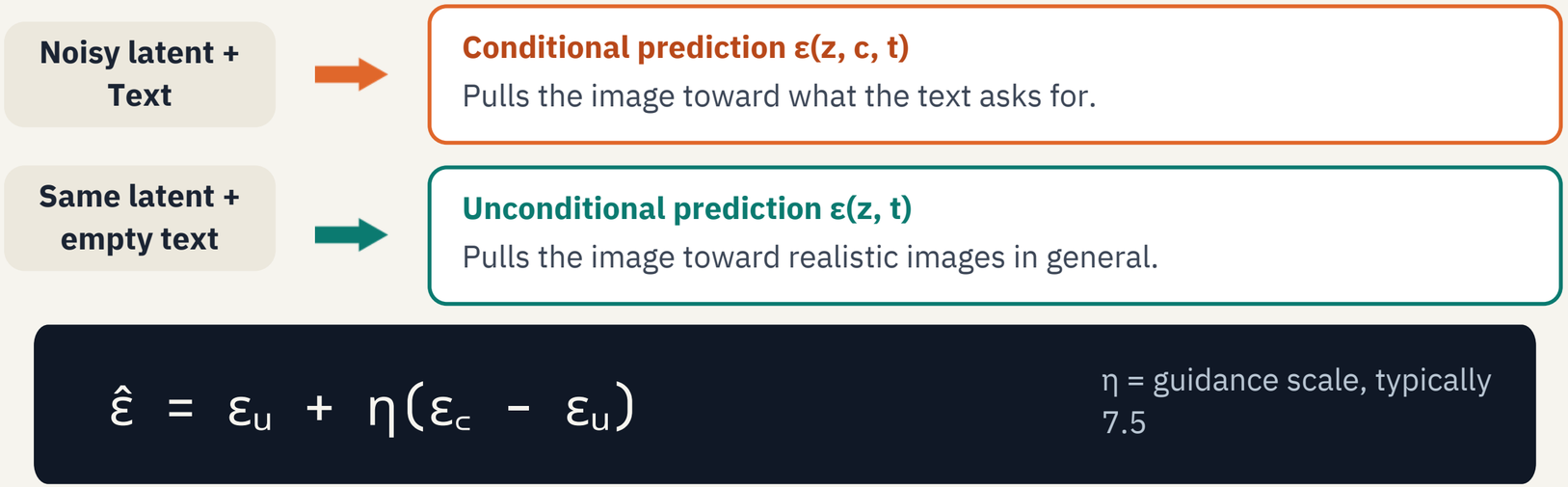
Introducing SafeGen

SafeGen modifies the model's visual representations to suppress explicit image generation, even under adversarial prompts.



METHOD · PREREQUISITE

Understanding Classifier-Free Guidance



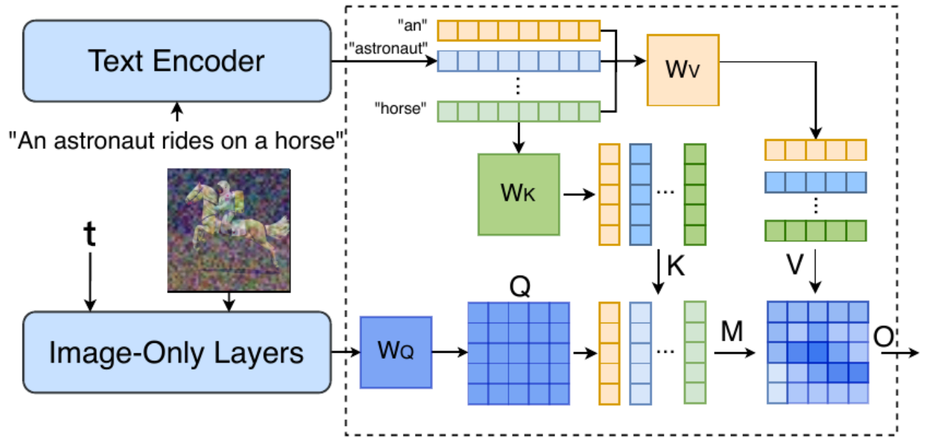
$$\hat{\varepsilon} = \varepsilon_u + \eta(\varepsilon_c - \varepsilon_u)$$

η = guidance scale, typically 7.5

Both predictions use the same U-Net weights. SafeGen edits vision only self-attention through the unconditional process, affecting image generation even when text prompts change.

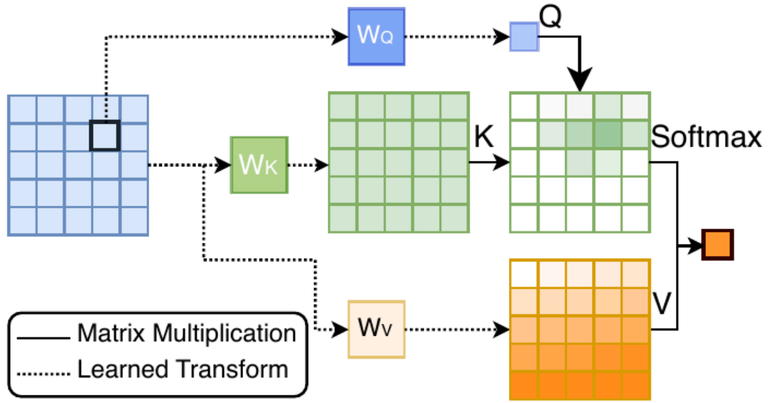
Cross-Attention vs. Self-Attention

Cross-attention: text meets image



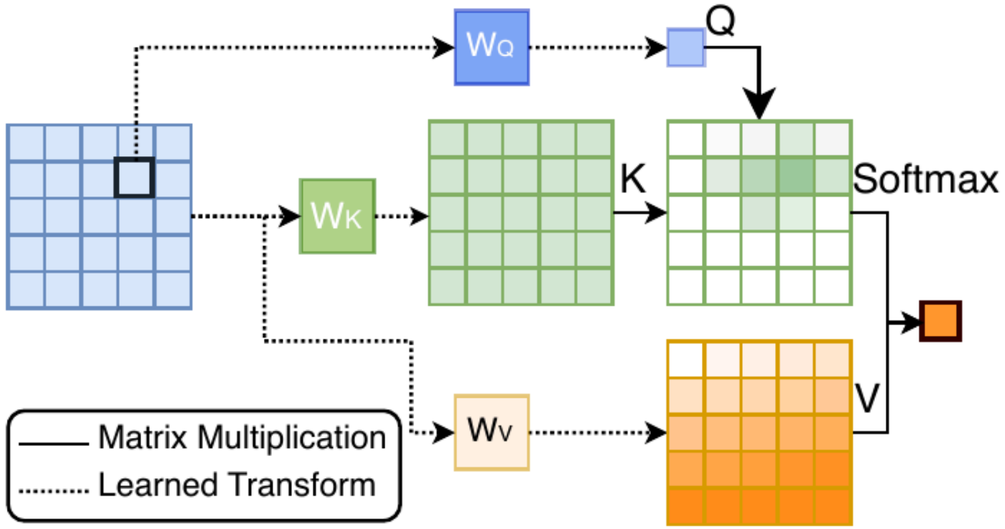
Query Q comes from the image latent while the keys K and values V come from the text tokens. This is where the prompt steers the picture.

Self-attention: image looks at itself



Q, K and V all come from the same image latent. Each region weighs its relation to every other region. No text is involved.

Why Does SafeGen Modify Self-Attention?



Paper Fig. 4: Diagram of Self-Attention

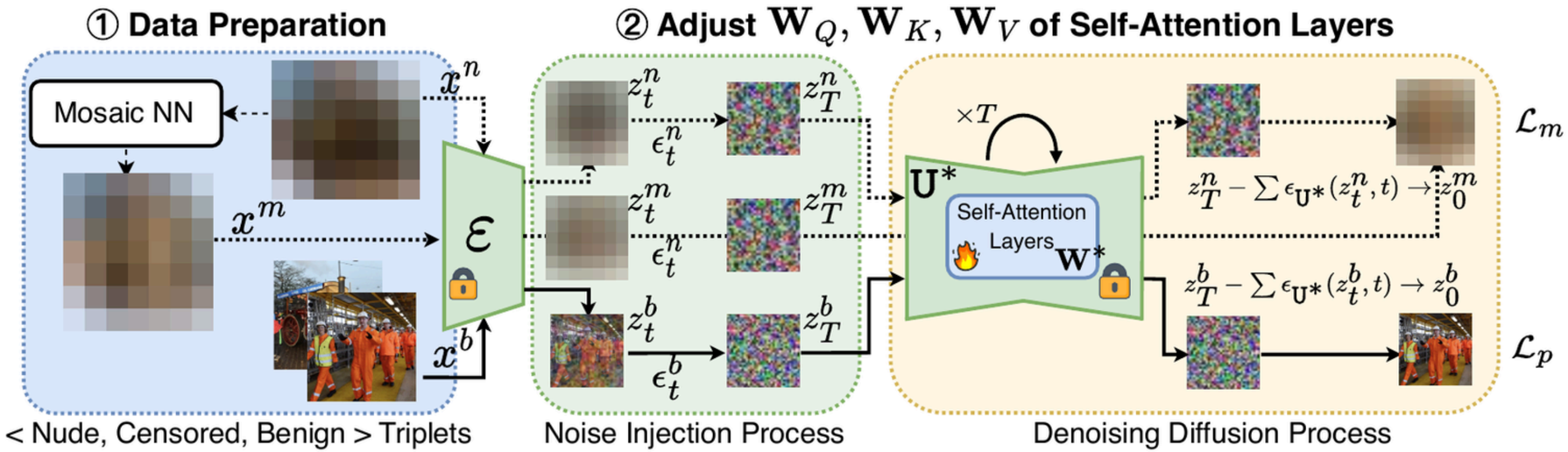
What the experiment shows

- Same text guidance in both rows.
- Only the unconditional process changes.
- Generated images become distorted or unrecognizable.

Why self-attention, not convolutions

- Captures long-range relationships between visual regions.
- Models global visual structure and different object scales.
- Can be edited without changing text-dependent cross-attention.

SafeGen Architecture: The Full Pipeline



1. Prepare image triplets
2. Encode and inject noise
3. Edit self-attention only
4. Apply two losses
5. Optimize and train

METHOD · STEP 1 OF 5

Step 1: Preparing the Training Data

Nude

x^n

Original explicit image used to teach the model what to suppress.

Mosaic

x^m

Censored version showing what the model should produce instead.

Benign

x^b

Normal image used to preserve the model's original behavior.

100 training triplets used to edit SafeGen’s self-attention weights.

An automated mosaic network creates censored targets from explicit images, removing the need to manually label sensitive regions

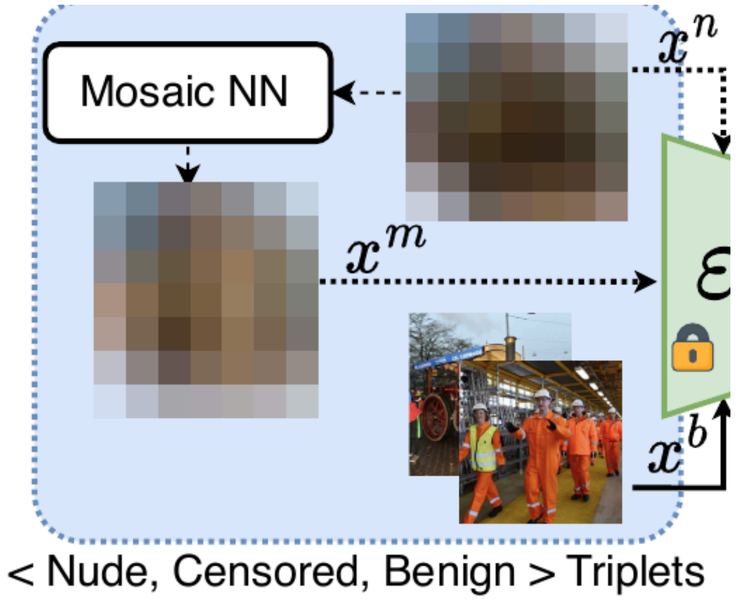


Fig. 8, data preparation (pixelated).

Step 2: Noise Injection and Text-Agnostic Training

a **Encode to latent space**
Convert all three images into compressed visual representations using the frozen image encoder.

b **Inject Noise into the Latents**
Add noise to simulate diffusion training. The nude and mosaic latents receive identical noise so their difference represents the content to censor.

c **Use Empty Text Conditioning**
Train with a blank prompt (""), allowing SafeGen to learn from visual information without relying on specific words.

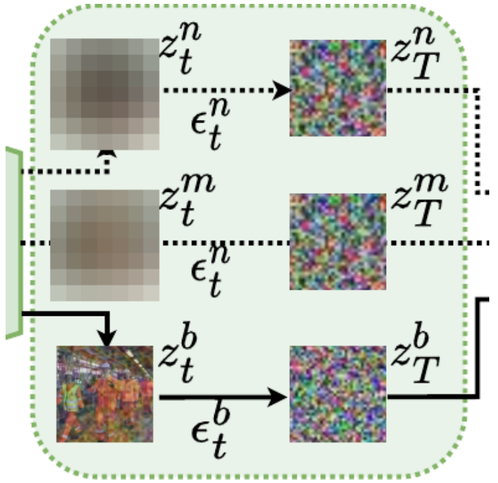


Fig. 8, noise injection.

Step 3: Editing the Self-Attention Layers

Image encoder and decoder	frozen
CLIP text encoder	frozen
Cross-attention layers	unchanged
Convolution blocks	unchanged
Self-attention: W_Q, W_K, W_V	**edited**

SafeGen updates only the self-attention weights (WQ, WK, WV) to suppress explicit visual representations. All other model parameters remain frozen to help preserve normal image generation.

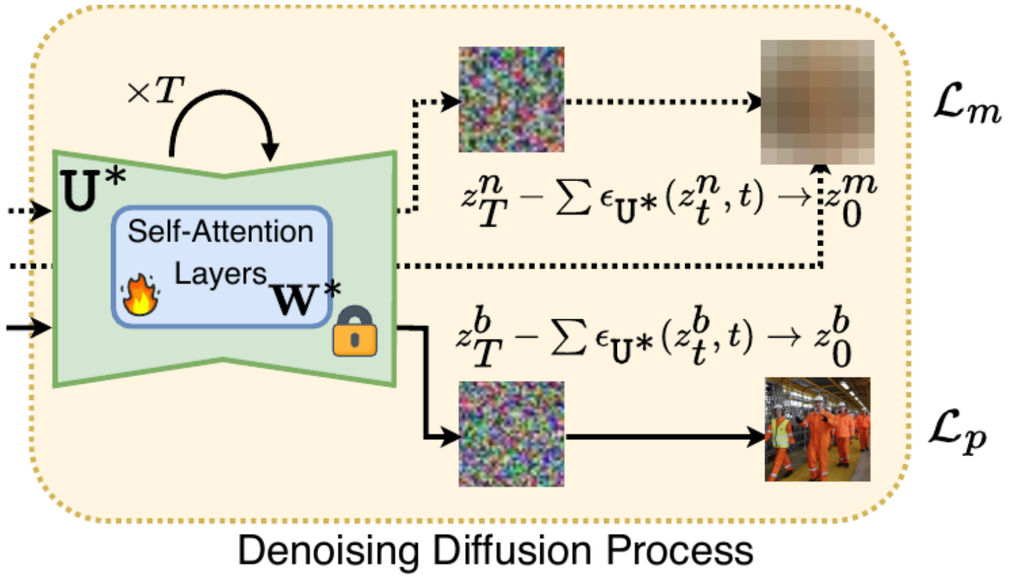


Fig. 8, denoising: flame = trainable, lock = frozen.

Step 4: SafeGen's Two Loss Functions

MOSAIC LOSS · L_M

Teach the model to censor explicit content

The U-Net learns to transform a noisy explicit image representation toward its mosaic-censored version.

Explicit latent → Mosaic latent

$$L_m = \Sigma \, \|\varepsilon*(z_t^n, t) - (z_T^n - z_T^m + \Sigma \, \varepsilon_t^n)\|^2$$

Suppress explicit visual features.

PRESERVATION LOSS · L_P

Teach the model to preserve benign content

The U-Net learns to remove added noise normally so benign images can still be reconstructed accurately.

Benign latent → Same Benign latent

$$L_p = \Sigma \, \|\varepsilon*(z_t^b, t) - \varepsilon_t^b\|^2$$

Maintain normal image quality.

The explicit image and its mosaic version receive identical noise, allowing SafeGen to isolate the visual differences that should be censored.

METHOD · STEP 5 OF 5

Step 5: Optimization and Training

$$\min W^* : \lambda_m \cdot L_m + \lambda_p \cdot L_p$$

$\lambda_m = 0.1$ | **$\lambda_p = 0.9$: most of the weight goes to preserving benign images**

AdamW	10^{-5}	1,000	100	200
optimizer	learning rate	training steps	image triplets	warm-up steps

Increasing the mosaic loss weight improved the emphasis on censorship but reduced benign image quality. The researchers selected $\lambda_m = 0.1$ and $\lambda_p = 0.9$ to balance safety with normal image generation.

METHOD · SYSTEM VIEW

Integration with Existing Defenses

Text-Dependent Defenses — SLD & ESD

$\varepsilon(z_t, c, t)$

SLD uses safety guidance to steer generation away from unsafe concepts, while ESD edits model weights to suppress those concepts.

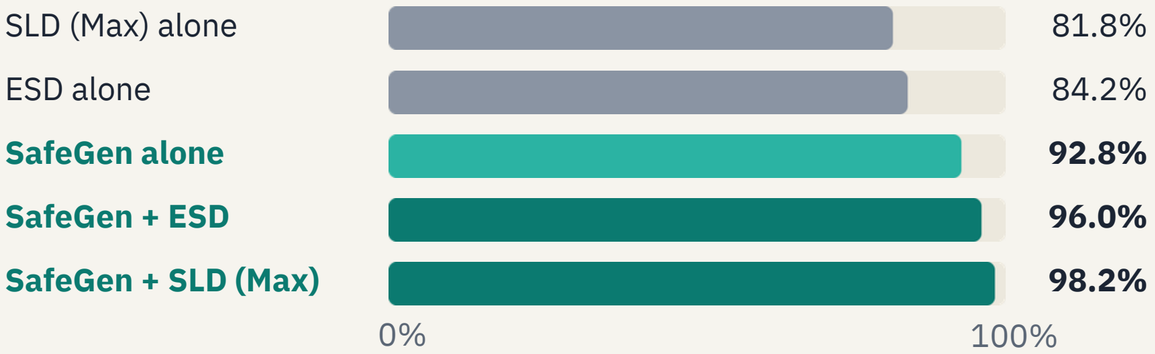
+

Vision-Based Defense — SafeGen

$\varepsilon(z_t, t)$

SafeGen edits self-attention weights using image-only training, reducing reliance on specific words or prompts.

Combining Defenses Improves Nudity Removal



Trade-off: Safety vs. Image Quality
Combining SafeGen with SLD (Max) achieves the highest nudity removal (98.2%), but reduces benign text-image alignment. Weaker SLD guidance better preserves benign generation quality.



Evaluation Check

Does it block unsafe images, and what does that cost?

EVALUATION

Experimental Setup and Evaluation Metrics

Stable Diffusion v1.4 as the base model · one NVIDIA A100-40GB GPU · PyTorch 1.12

DATA SETS

I2P
931 hand-written sexual prompts

SneakyPrompt
Optimized prompts: natural (N) and pseudo-word (P)

NSFW-56k
56,000 captions of real explicit images

COCO-25k
25,000 benign captions (the quality test)

METRICS

NRR
Nudity removed vs. SD v1.4. Higher is better.

CLIP score
Text-image match. Low on unsafe, high on benign.

LPIPS
Perceptual distance to a reference. Lower is better.

FID
Distance to real-image statistics. Lower is better.

BASELINES

Original SD · Safety filter · SD-V2.1 · ESD · SLD Weak · SLD Medium · SLD Strong · SLD Max

EVALUATION

Does SafeGen Prevent Unsafe Image Generation?

99.4%

best reported NRR, on the NSFW-56k dataset

97.1%

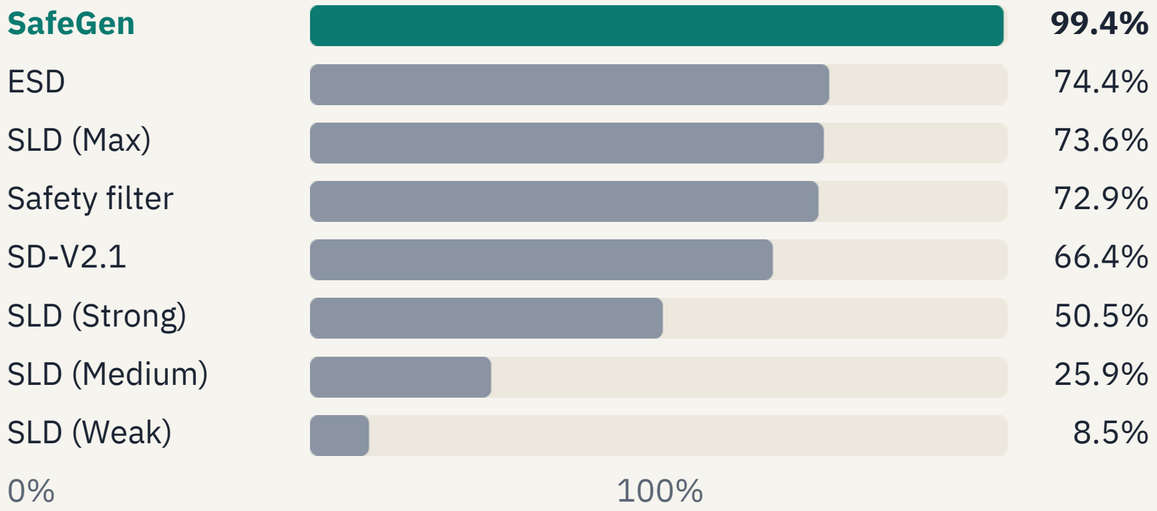
average NRR across the four adversarial test sets

98.2%	98.0%	92.7%	99.4%
Sneaky-N	Sneaky-P	I2P	NSFW-56k

Adversarial CLIP score: lower is safer

SafeGen 17.16 on NSFW-56k; the eight baselines range from 20.56 to 26.61. SafeGen is lowest on all four sets (Table 2).

NRR on NSFW-56k by method (Table 1)



SafeGen substantially outperformed every comparison defense on this benchmark. Exposed body parts detected on NSFW-56k: **4,533** with the original model, **27** with SafeGen (Fig. 10).

EVALUATION

Does SafeGen Preserve Benign Image Quality?

Method	CLIP	LPIPS	FID-25k
Original SD	24.56	0.782	20.05
SD-V2.1	24.53	0.777	18.27
ESD	23.97	0.788	20.36
SLD (Max)	23.03	0.801	27.57
SLD (Strong)	23.57	0.792	25.17
SLD (Medium)	24.17	0.786	23.19
SLD (Weak)	24.57	0.783	20.24
SafeGen (ours)	24.33	0.787	20.31

Benign prompts from COCO-25k (Table 3). The safety filter is left out because it does not change benign images.

CLIP: higher is better
Image matches the prompt. 24.33 vs. 24.56 for the original.

LPIPS: lower is better
Looks like the reference photo. 0.787 vs. 0.782.

FID: lower is better
Overall realism of the image set. 20.31 vs. 20.05.

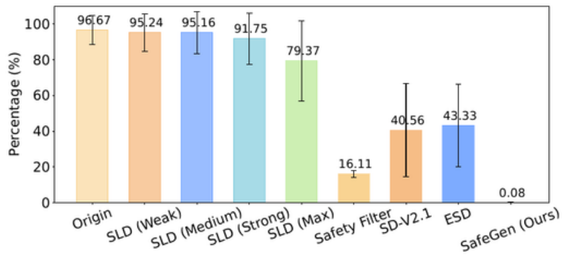
Near-original quality. The strongest text-based setting, SLD (Max), pays for safety with an FID of 27.57.

EVALUATION

Human Evaluation: Does SafeGen Actually Work?

82 adult participants (aged 21 to 47; 53 male, 29 female) answered a five-part, IRB-approved questionnaire.

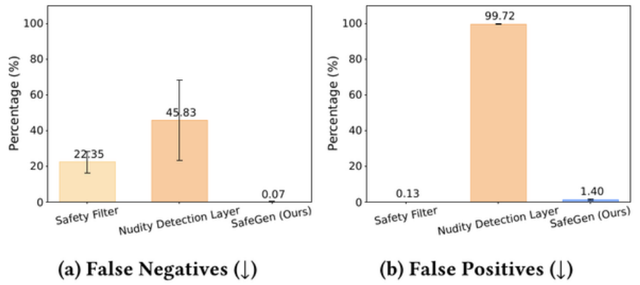
Is the output still explicit?



0.08%

of SafeGen images judged explicit. Original: 96.67%. Safety filter: 16.11%.

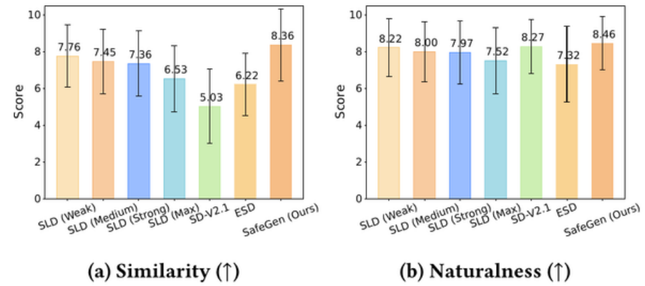
What slips through?



0.07%

false negatives, vs. 22.35% for the safety filter. False positives: 1.40%.

Do benign images hold up?



8.36 / 10

similarity to the original model's images, the top score. Naturalness: 8.46.

PART 6 OF 6



Future Work

What the paper leaves open.

FUTURE WORK · RAISED BY THE AUTHORS

Limitations and Future Research Directions

Nudity is only a proxy

NRR counts exposed body parts, but explicit is not the same as nude. Users still rated 79% of SLD (Max) images as explicit.

Risk of over-censorship

Non-explicit nudity, such as a nude sculpture, can be mosaicked too. Adding such images to the benign set helps.

Cultural variation

What counts as sexually explicit differs between societies. A clear standard and better objective metrics are still missing.

Extending the idea

Apply vision-only editing to image-to-image and text-to-video models, and integrate SafeGen into the Diffusers library.

PRESENTER'S ANALYSIS

Critical Analysis: What Remains Unproven?

Access to model weights

Could an attacker replace the model or fine-tune away SafeGen's safety edits?

Adaptive adversarial attacks

How would SafeGen perform against prompts optimized specifically to bypass SafeGen?

Limited generalizability

Evaluated primarily on Stable Diffusion v1.4 and one unsafe-content category.

Reliance on proxy metrics

NudeNet measures detected nudity, not every form of sexually explicit content.

CONCLUSION

Questions?

